Anupa Supul

Computer Science Undergraduate

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Summary

Computer Science undergraduate at the University of Ruhuna and aspiring Full Stack Developer with experience in full-stack development, backend engineering, AI/ML research, and workflow automation. Experienced in building web applications using React.js, Next.js, Node.js, and Spring Boot, developing RAG-based AI applications, and working with AWS and Docker. Co-authored research publications on rice leaf disease detection. Passionate about building practical, scalable software solutions and exploring AI and cloud technologies.

Education

University of Ruhuna *September 2023–Present*

Bachelor of Computer Science

St. Thomas' Boys' College, Matara

- G.C.E. Advanced Level – Physical Science
- Combined Mathematics – A Chemistry – B Physics – C Z-Score: 1.4847

Professional Experience

BotCalm — Intern Software Engineer *April 2026–October*

- Assisting in the development of web applications using modern technologies.
- Working with React.js, Next.js, REST APIs, and database-driven applications.
- Collaborating with team members in an agile software development environment.
- Gaining practical experience in full-stack software development and software engineering workflows.

Technical Skills

Programming Languages: Java, JavaScript, Python, Go (Basic)

Frontend: React.js, Next.js, Tailwind CSS

Backend & APIs: Node.js, Express.js, Spring Boot, FastAPI, REST APIs, JWT Authentication

Databases & Retrieval: MongoDB, MySQL, ChromaDB, Vector Search, Semantic Embeddings

AI/ML: TensorFlow, Keras, CNN,, VGG19, RAG, LLM API Integration, Sentence Transformers

Cloud & DevOps: AWS (IAM, S3, EC2, VPC, ECR, ECS, Fargate, ALB, Secrets Manager, CloudWatch), Docker, Docker Compose, GitHub Actions, CI/CD

Automation & Integration: n8n, Workflow Automation, Gemini API, OpenRouter API, ElevenLabs, Creatomate, YouTube API

Tools: Git, GitHub, Postman, Trello, Figma

Certifications

AWS Cloud Practitioner Essentials | AWS Technical Essentials | AWS Cloud Quest: Cloud Practitioner

Projects

MediCare — Hospital Management System(Final Year Group Project) *October 2025–January*

Technologies: Spring Boot, React.js, Tailwind CSS, JWT Authentication, MySQL, Gemini API

- Developed a scalable hospital management system to manage patients, appointments, and staff workflows.
- Designed and implemented backend services using Spring Boot with microservices architecture.
- Implemented secure authentication and authorization using JWT.
- Built responsive frontend interfaces using React.js and Tailwind CSS.
- Integrated Google Gemini API for AI-powered chatbot assistance.
- Integrated a pre-trained ECG machine learning model and implemented rule-based logic for automated health data analysis and intelligent alerts.
- Used MySQL for secure data management and GitHub/Trello for collaboration and project tracking.

Sri Lankan Legal RAG — AI-Powered Legal Research Assistant *October 2026–Present*

Technologies: Python, FastAPI, React.js, ChromaDB, Sentence Transformers, OpenRouter API, Docker, Docker Compose, GitHub Actions, Amazon Web Services (AWS)

- Developed a multilingual-ready Retrieval-Augmented Generation (RAG) legal assistant grounded in Sri Lankan legislation, including the Constitution, Inland Revenue Act, and intellectual property legislation.
- Implemented hybrid retrieval combining semantic embeddings, vector search, keyword matching, and legal-concept scoring to find relevant legal provisions.
- Integrated the OpenRouter LLM API to generate context-aware answers grounded in retrieved evidence, with source references and confidence-based abstention when evidence is insufficient.
- Built REST APIs with FastAPI and a React.js frontend, using Sentence Transformers and ChromaDB for semantic embeddings and persistent vector storage.
- Containerized frontend and backend services with Docker and Docker Compose; implemented GitHub Actions CI/CD for validation, image builds, and deployment.
- Deployed containerized services on AWS using Amazon ECR, ECS/Fargate, a VPC, and an Application Load Balancer; integrated S3, AWS Secrets Manager, and CloudWatch for storage, secrets, logging, and monitoring.

AI-Powered YouTube Content Automation Workflow *April2026–June*

Technologies: n8n, Gemini API, ElevenLabs, Creatomate, YouTube API

- Developed an end-to-end automated content generation and publishing workflow using n8n.
- Integrated Gemini AI for automated script generation.
- Integrated ElevenLabs for AI-powered voice synthesis.
- Automated video rendering using Creatomate APIs.
- Implemented automated YouTube video publishing through API integrations.
- Designed a scalable workflow that minimizes manual effort through automation.
- Published the project on GitHub with documentation and workflow architecture.

Maths Class Management System

Combined Mathematics – Mr. Gayan Gammanavidana *September 2025–Present*

Technologies: MERN Stack (MongoDB, Express.js, React.js, Node.js)

- Developed a full-stack class management system for a Combined Mathematics instructor.
- Implemented QR code generation and scanning to automate student attendance and payment verification.
- Built modules for attendance tracking, payment management, and student records.
- Designed responsive user interfaces using React.js for administrative and operational workflows.
- Collaborated directly with the client to gather requirements and refine system features.
- Currently in the final deployment phase.

Rice Plant Disease Detection(Research based) — Web Application *December 2025–Present*

Technologies: React.js, Tailwind CSS, Spring Boot, MySQL, REST APIs, TensorFlow/Keras, VGG19

- Developed a research-oriented rice leaf disease detection web application.
- Built an end-to-end system including a responsive frontend and backend API.
- Trained and evaluated CNN-based models to improve detection accuracy and reliability.
- Implemented a practical solution for agricultural disease identification.

Rice Leaf Disease Detection Research *November 2024–Present*

- Conducted deep learning research to classify rice leaves as healthy or diseased using image-based detection.
- Created and curated datasets for training and testing.
- Applied data preprocessing and augmentation techniques to improve model generalization.
- Implemented and compared VGG16, VGG19, Custom CNN, and CNN+SVM architectures.
- Achieved 95.27% validation accuracy and 92.62% test accuracy for brown spot disease detection.

Research Publications

Identification of Brown Spot Leaf Disease Using Machine Learning *November 2025*

APSURS 2025 – Sabaragamuwa University of Sri Lanka

- Published research paper at the 4th Applied Sciences Undergraduate Research Symposium (APSURS 2025).

Machine Learning and Deep Learning-Based Approaches for Rice Leaf Disease Detection in Sri Lanka *February 2026*

RITSCON 2026 – University of Ruhuna

- Published and presented research findings at RITSCON 2026.

Upcoming Publications and Projects

Rice Plant Leaf Disease Image Dataset *Planned Release: December 2026*

- Currently developing and preparing a rice leaf image dataset captured under real-world conditions for machine learning and deep learning research.

Volunteering & Community Involvement

- Organizing Committee Member & Public Relations Team Leader — RushCoder 3.0 Hackathon, University of Ruhuna
- Member — Sasnaka Sansadaya

References

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